

Chapter 2

The Log-Quantile-Density Model of the Volatility Smile

A volatility smile is a probability law viewed through the Black formula. The log-quantile-density model builds the law directly: start from percentile rank, move through log-return at a strictly positive speed, then shift the result to match the forward. Two tail exponents span the range from exponential to Gaussian-rate log-return tails. One cumulative upper-share function then supplies prices, digitals, density, local smile derivatives, moments, calibration sensitivities, and calendar constraints. Across expiries, ordering that function on the whole rank line is exactly equivalent to noncrossing total-variance curves on the whole strike line.

2.1 From a volatility smile to a probability law

A listed option board contains finitely many strikes. A pricing system needs a curve at every strike, including strikes that are not quoted. Traders view that curve as a volatility smile. Mathematically it is the option-price image of a probability distribution. The two views are equivalent, but the second one makes the restrictions clear: option prices must decrease with strike, butterflies must have nonnegative value, the terminal asset price must remain positive, and its mean must equal the forward.

The LQD model starts from this distribution rather than from volatility. Its central device is a monotone map from percentile rank to log-return. A positive speed makes the map increasing and therefore makes the density positive. The smile is calculated only after the law has been built.

The conceptual core is contained in sections 2.1 to 2.4: law, transport, tails, and pricing. Sections 2.5 to 2.7 turn those identities into a fitted surface. The proofs and implementation details are collected in the chapter appendices so that the main argument can be read without interrupting its progression.

2.1.1 Normalize first

Fix an expiry T . Let F be its forward and D its discount factor. Work under the T -forward measure and set

$$X = \log \frac{S_T}{F}, \quad Y = e^X = \frac{S_T}{F}, \quad \mathbb{E}[Y] = 1. \quad (2.1)$$

The expectation may simply be read as the normalized pricing expectation; no dynamic model is needed in this chapter. The last equality is the forward, or martingale, condition. For a strike K write

$$k = \log(K/F), \quad y = e^k = K/F.$$

Writing $a^+ = \max(a, 0)$, the normalized call and put prices are

$$c(k) = \mathbb{E}[(Y - e^k)^+], \quad P(k) = \mathbb{E}[(e^k - Y)^+], \quad c(k) - P(k) = 1 - e^k. \quad (2.2)$$

Multiplying by DF returns the cash price. This normalization removes rates, dividends, and currency units from the one-expiry problem.

Markets usually quote implied volatility rather than normalized price. If τ is the chosen year fraction and $w(k) = \sigma(k)^2\tau$ is total implied variance, the normalized Black call is

$$B(k, w) = \Phi(d_+) - e^k \Phi(d_-), \quad d_{\pm} = -\frac{k}{\sqrt{w}} \pm \frac{\sqrt{w}}{2}. \quad (2.3)$$

For every admissible price strictly between intrinsic value and the forward, the implied variance $w(k)$ is the unique positive solution of $B(k, w(k)) = c(k)$ [2]. Price and implied volatility are therefore two coordinates for the same object.

2.1.2 The call curve and the law

Use normalized strike y , not log-strike k , for the next calculation. If Y has density f_Y , differentiating the call payoff gives

$$\frac{\partial c}{\partial y} = -\mathbb{P}(Y > y), \quad \frac{\partial^2 c}{\partial y^2} = f_Y(y) \geq 0. \quad (2.4)$$

This is the Breeden–Litzenberger identity [3]. The slope of the call curve is minus the probability of finishing above strike; its curvature is the terminal density.

The same identity has a traded reading. For a strike spacing δ , the symmetric butterfly

$$c(y - \delta) - 2c(y) + c(y + \delta)$$

has a nonnegative payoff in every terminal state. Dividing its price by δ^2 and shrinking δ recovers $f_Y(y)$. Negative curvature would give a negative price to a nonnegative payoff.

Definition 2.1 (Arbitrage-free slice). An arbitrage-free slice is a call curve generated by a positive random variable Y with $\mathbb{E}[Y] = 1$. If Y has no atom at zero and has a density, the curve is equivalently a decreasing convex function of $y > 0$ satisfying

$$(1 - y)^+ \leq c(y) \leq 1, \quad -1 \leq c'(y) \leq 0, \quad c(y) \rightarrow 0 \quad (y \rightarrow \infty),$$

with $c'(y) \rightarrow -1$ as $y \downarrow 0$.

Every condition has a direct source. The upper bound follows from $(Y - y)^+ \leq Y$; the lower bound is Jensen's inequality; the slope is a survival probability; and the right boundary follows from integrability of Y . Conversely, the curvature of such a curve defines a probability measure with mean one. Thus a valid call curve and a positive mean-one law contain exactly the same information.

One distinction matters in practice: convexity belongs to strike y , not to log-strike k . The chain rule gives

$$\frac{\partial^2 c}{\partial k^2} = e^k \{e^k f_Y(e^k) - \mathbb{P}(Y > e^k)\},$$

which is negative far in the left wing for an ordinary density. Testing convexity in log-strike rejects correct curves.

In volatility coordinates the same requirement becomes less transparent. Twice differentiating $c(k) = B(k, w(k))$ yields

$$f_X(k) = g_D(k) \frac{\varphi(d_-(k, w(k)))}{\sqrt{w(k)}}, \quad g_D(k) = \left(1 - \frac{kw'}{2w}\right)^2 - \frac{(w')^2}{4} \left(\frac{1}{w} + \frac{1}{4}\right) + \frac{w''}{2}. \quad (2.5)$$

A twice differentiable smile is butterfly-free exactly when $g_D(k) \geq 0$ for every k [8]; the calculation is given in section 2.A.1. Direct smile models must control this nonlinear expression. LQD instead constructs $f_X > 0$, so the condition is a consequence rather than a calibration constraint.

2.1.3 Four coordinates, used for four different jobs

The chapter changes horizontal coordinate several times. Each change has a specific purpose:

coordinate	meaning	main use
$y = K/F$	normalized strike	convexity and static arbitrage
$k = \log y$	log-moneyness	market smiles
$u \in (0, 1)$	percentile rank	probability and quantiles
$z = \log\{u/(1-u)\}$	log-odds	integration and numerics

The logit map $u \mapsto z$ is a bijection from $(0, 1)$ to $\mathbb{R}$. Its inverse and the logistic density are

$$u = \Lambda(z) = \frac{1}{1 + e^{-z}}, \quad \rho(z) = \Lambda(z)\Lambda(-z). \quad (2.6)$$

The median is $z = 0$; the tenth and ninetieth percentiles are $z = \mp \log 9$. Moving one unit in z multiplies the odds by e . This turns the open rank interval into an ordinary real line and makes both tails numerically accessible.

The complete construction follows one direction:

$$\text{rank } u \longrightarrow \text{log-odds } z \longrightarrow \text{return } x(z) \longrightarrow \text{law } f_X \longrightarrow \text{price } c \longrightarrow \text{smile } w.$$

Rank supplies exactly one unit of probability. Positive transport speed makes the return map increasing. The Black formula is used only at the final change of coordinates. This order is the organizing principle of the chapter.

2.2 The model: stretch probability rank

2.2.1 Quantiles and transport speed

Let $Q(u)$ be the quantile function of X . If U is uniform on $(0, 1)$, then $Q(U)$ has that law. Write $q(u) = Q'(u)$ for the quantile density. Differentiating $F_X(Q(u)) = u$ gives

$$f_X(Q(u)) = \frac{1}{q(u)}. \quad (2.7)$$

The derivative q measures log-return distance per unit of percentile rank. Small q compresses many ranks into a narrow return interval and creates high density; large q spreads the same probability over a wider interval and creates low density. A strictly positive q makes Q strictly increasing. Parameterizing $\log q$ makes that positivity automatic.

The original LQD model starts from

$$q(u) = \frac{e^{g(u)}}{u(1-u)}. \quad (2.8)$$

The factors $u(1-u)$ make q diverge at both endpoints, giving unbounded support. They also cancel the derivative of the logistic map. Since $du/dz = u(1-u)$,

$$\frac{dQ}{dz} = e^{g(\Lambda(z))}. \quad (2.9)$$

Thus g is the logarithm of the speed at which log-return moves along the log-odds line. If g tends to a constant in a tail, that speed tends to a positive constant and the log-return tail is exponential.

To permit lighter tails, slow the transport gradually at extreme ranks. Introduce the positive logarithmic gauges

$$\ell_-(u) = 1 - \log u, \quad \ell_+(u) = 1 - \log(1-u). \quad (2.10)$$

They behave like $\log(1/u)$ on the left and $\log(1/(1-u))$ on the right. Choose tail exponents $\alpha_-, \alpha_+ \in [0, 1/2]$ and set

$$\log q(u) = -\log u - \log(1-u) - \alpha_- \log \ell_-(u) - \alpha_+ \log \ell_+(u) + g(u). \quad (2.11)$$

Equivalently,

$$q(u) = \frac{e^{g(u)}}{u(1-u)\ell_-(u)^{\alpha_-}\ell_+(u)^{\alpha_+}}. \quad (2.12)$$

This replaces the left endpoint factor u by $u\{\log(1/u)\}^{\alpha_-}$ to leading order, and does the analogous thing on the right.

Because $du/dz = u(1-u)$, the two leading endpoint singularities cancel:

$$\frac{dQ}{dz} = \frac{e^{g(\Lambda(z))}}{\ell_-(\Lambda(z))^{\alpha_-}\ell_+(\Lambda(z))^{\alpha_+}}. \quad (2.13)$$

The endpoint singularities have disappeared. The distribution is obtained by integrating an ordinary positive function on the real line. Setting $\alpha_- = \alpha_+ = 0$ recovers the original model. Increasing one exponent slows the corresponding extreme transport and lightens that tail.

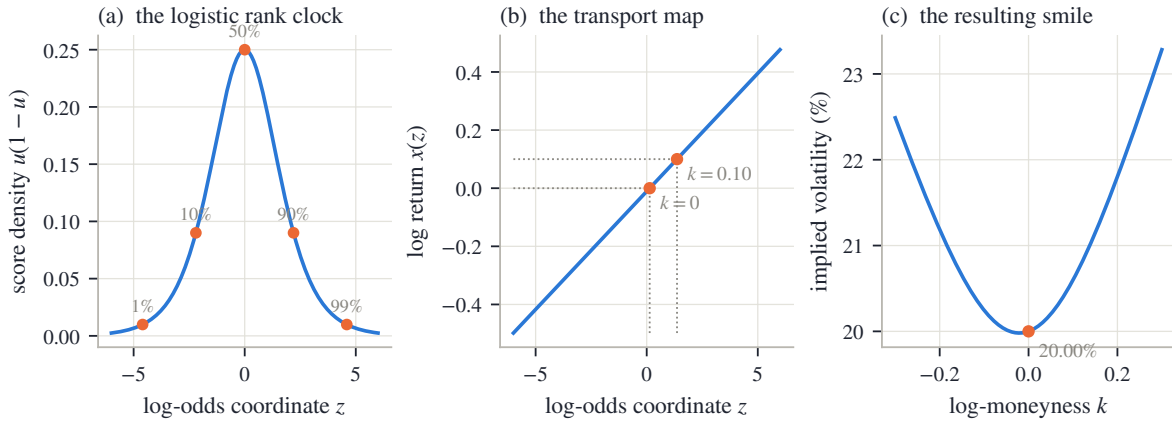


Figure 2.1: The transport construction in the exponential-tail subclass with constant speed. Left: logistic log-odds and five percentile marks. Center: an increasing map from log-odds to log-return; strike roots are read on this map. Right: the implied-volatility smile obtained by pricing the resulting law. The first two panels define the model; the last changes coordinates.

2.2.2 A complete slice

The function g controls the body of the law. A low-degree polynomial is enough for smooth smiles and keeps the number of effective shapes visible. Shifted Legendre polynomials are orthogonal under uniform rank, so their coefficients are substantially better scaled than raw powers of u .

Definition 2.2 (Generalized LQD slice). Fix $N \geq 2$, tail exponents $\alpha_{\pm} \in [0, 1/2]$, and coefficients $\theta = (L, R, a_2, \dots, a_N)$. Define

$$g(u) = (1-u)L + uR + \sum_{n=2}^N a_n P_n(1-2u). \quad (2.14)$$

$$x'(z) = \frac{e^{g(\Lambda(z))}}{\ell_-(\Lambda(z))^{\alpha_-}\ell_+(\Lambda(z))^{\alpha_+}}. \quad (2.15)$$

$$\boxed{X = x(Z), \quad x(z) = m + \int_0^z x'(t) dt, \quad Z \sim \Lambda.} \quad (2.16)$$

The shift m is chosen, when possible, so that $\mathbb{E}[e^X] = 1$.

For every finite coefficient vector, $x'(z) > 0$. Since $\alpha_{\pm} < 1$, the integral of x' diverges at both ends, so x is a smooth increasing bijection from $\mathbb{R}$ to $\mathbb{R}$. Consequently

$$F_X(x(z)) = \Lambda(z), \quad f_X(x(z)) = \frac{\rho(z)}{x'(z)} > 0. \quad (2.17)$$

The map cannot fold, stop, or reverse. The density is positive before any market quote is fitted. The parameters now have three distinct roles:

- $\alpha_{\pm}$ select the two asymptotic tail classes;
- the endpoint values of g set the two tail scales;
- the remaining coefficients shape the distribution between the tails.

The endpoint coefficients are

$$\lambda_- = e^{g(0)} = \exp\left(L + \sum_{n \geq 2} a_n\right), \quad \lambda_+ = e^{g(1)} = \exp\left(R + \sum_{n \geq 2} (-1)^n a_n\right). \quad (2.18)$$

When $\alpha_{\pm} = 0$, these numbers are the limiting transport speeds. For $\alpha_{\pm} > 0$, the speeds tend to zero and $\lambda_{\pm}$ multiply their power-law decay in $|z|$. In both cases $\lambda_{\pm}$ are tail scales, while $\alpha_{\pm}$ determine the type of tail.

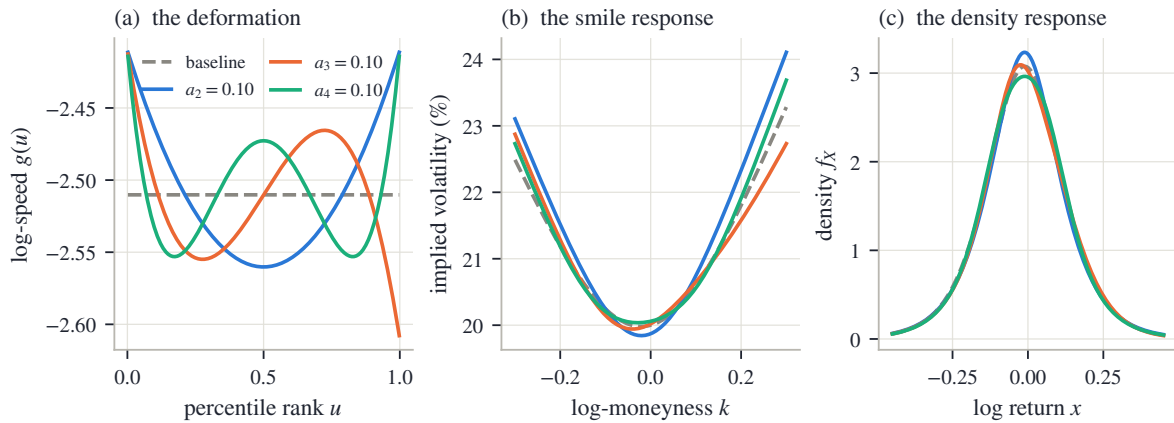


Figure 2.2: Three Legendre perturbations of one exponential-tail reference law. Left: their action on the log-speed g . Center: the implied-volatility response. Right: the density response. Raising g stretches return space and lowers density at the affected ranks. Even modes act on both sides; odd modes transfer shape from one side to the other.

Raw Legendre coefficients move both the body and the endpoint scales. That mixing is inconvenient in calibration. Replace them by the endpoint-vanishing modes

$$\tilde{P}_n(u) = P_n(1 - 2u) - (1 - u) - (-1)^n u, \quad n \geq 2. \quad (2.19)$$

They satisfy $\tilde{P}_n(0) = \tilde{P}_n(1) = 0$. The equivalent parameter vector $(\log \lambda_-, \log \lambda_+, a_2, \dots, a_N)$ lets an optimizer change body shape without silently changing the asymptotic tails.

2.2.3 One shift matches the forward

The transport fixes the shape of the log-return law but leaves its location free. Translating every log-return by the same number changes neither ranks, density shape, nor tail class. Use this one degree of freedom to impose $\mathbb{E}[e^X] = 1$.

Let

$$\bar{x}(z) = \int_0^z x'(t) dt, \quad I = \int_{-\infty}^{\infty} e^{\bar{x}(z)} \rho(z) dz.$$

Proposition 2.3 (The forward condition). *If $I < \infty$, the unique martingale shift is $m = -\log I$. Moreover,*

$$I < \infty \iff \alpha_+ > 0 \text{ or } (\alpha_+ = 0 \text{ and } \lambda_+ < 1). \quad (2.20)$$

Proof. Since $X = m + \bar{x}(Z)$, $\mathbb{E}[e^X] = e^m I$, which gives the unique shift. If $\alpha_+ > 0$, then $\bar{x}(z) = O(z^{1-\alpha_+}) = o(z)$ while $\rho(z) \sim e^{-z}$, so the right integrand is integrable. If $\alpha_+ = 0$, then $\bar{x}(z) = \lambda_+ z + O(1)$ and the integrand is asymptotic to a constant times $e^{(\lambda_+-1)z}$. The left integrand is always integrable because $\bar{x}(z) \leq 0$ for $z \leq 0$ and $\rho(z) \sim e^z$. $\square$

Only the right tail can make the forward infinite. In the exponential-tail subclass its scale must satisfy the exact wall $\lambda_+ < 1$. Every strictly lighter right tail has a finite forward for any finite λ_+ . The jump at $\alpha_+ = 0$ is mathematical, not conventional: any positive exponent eventually beats a linear exponential rate. The convergence is slow when the exponent is small, which matters numerically in section 2.6.

2.2.4 Two reference laws anchor the model

Set $\alpha_{\pm} = 0$, $L = R = \log s$, and $a_n = 0$, with $0 < s < 1$. Then $x(z) = m + sz$ and X is logistic with scale s . Its moment-generating function is

$$\mathbb{E}[e^{tZ}] = \int_0^1 \left(\frac{u}{1-u} \right)^t du = \Gamma(1+t)\Gamma(1-t) = \frac{\pi t}{\sin(\pi t)}, \quad |t| < 1. \quad (2.21)$$

Hence

$$m = -\log \frac{\pi s}{\sin(\pi s)}, \quad \text{Var}(X) = \frac{\pi^2 s^2}{3}. \quad (2.22)$$

Every quantity is explicit, so this law is the primary end-to-end test of an implementation: transport, normalization, moments, strike roots, prices, and densities must all agree with it.

At the lighter endpoint of the tail range, a normal law $X \sim N(-s^2/2, s^2)$ has

$$q(u) \sim \frac{s}{u\sqrt{2\log(1/u)}} \quad (u \downarrow 0),$$

and the symmetric right relation. With $\alpha_{\pm} = 1/2$, the function g obtained from (2.11) has finite endpoint limits. Polynomial LQD log-speeds can therefore approximate the normal law without forcing g to diverge at either endpoint. The logistic and normal laws are not competing calibrations; they are exact benchmarks for the two ends of the tail range developed next.

2.3 What the construction guarantees, and what the tails choose

The construction separates properties that are often mixed together in a smile fit. Positive density is automatic for every finite coefficient vector. A finite forward imposes one exact restriction on an exponential right tail. The two tail exponents are modeling choices: they decide the moment domain and the continuation of the smile beyond observable strikes.

2.3.1 One-expiry no-arbitrage

Proposition 2.4 (Validity of one slice). *Let the parameters satisfy (2.20), and choose the shift of proposition 2.3. Then:*

- (i) X has a strictly positive continuous density on $\mathbb{R}$; $Y = e^X$ has a strictly positive density on $(0, \infty)$ and mean one;
- (ii) the call is strictly decreasing and strictly convex in normalized strike y , obeys the bounds in definition 2.1, and satisfies put–call parity;

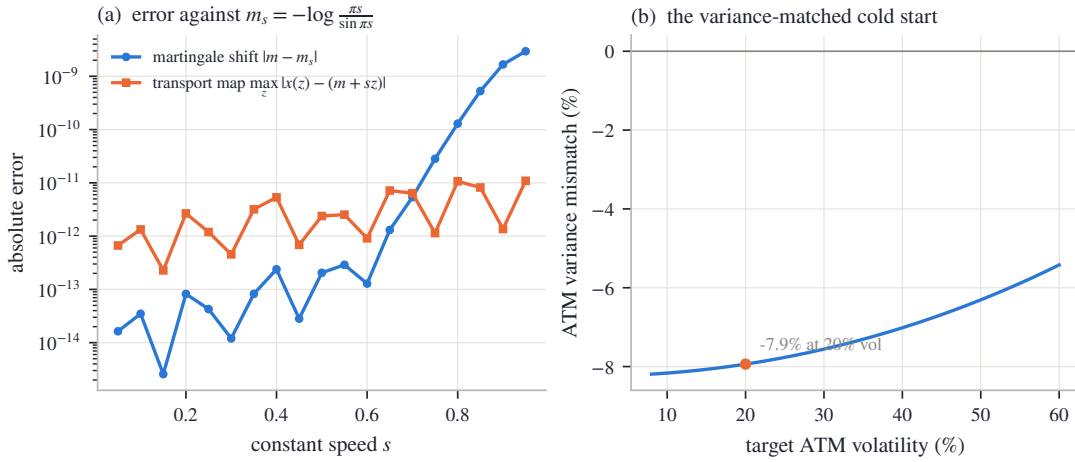


Figure 2.3: The constant-speed logistic slice as an implementation test. Left: computed transport and martingale shift errors against (2.22). Right: the variance-matched cold start and its small-variance ATM bias. The test exercises transport integration, normalization, tail treatment, and strike inversion with known answers.

(iii) every nondegenerate finite-width butterfly has positive price;

(iv) the implied-variance curve satisfies $g_D(k) > 0$ at every finite strike.

Proof. Equation (2.17) gives a positive continuous density and proposition 2.3 gives unit mean. In rank coordinates,

$$c(y) = \int_0^1 (e^{Q(u)} - y)^+ du.$$

Each integrand is decreasing and convex in y . Differentiation gives $c'(y) = -\mathbb{P}(Y > y)$ and $c''(y) = f_Y(y) > 0$. Jensen's inequality and dominated convergence give the boundary conditions. The butterfly and parity statements follow. Finally, (2.5) expresses the positive density as a positive Black factor times g_D . $\square$

This is a statement about the entire continuum of strikes, not merely the calibration grid. The numerical representation must preserve the same geometry; section 2.6.3 gives the corresponding checks.

2.3.2 What the tail exponents mean

The log-odds coordinate has a useful probabilistic meaning in the right tail: $\mathbb{P}(Z > z) \sim e^{-z}$. Hence z is asymptotically the negative logarithm of tail probability. If log-return grows linearly with z , its tail is exponential. If it grows like $\sqrt{z}$, its tail has Gaussian rate. The exponent α moves continuously between these two growth laws.

At the right endpoint, $1 - \Lambda(z) \sim e^{-z}$, $\ell_+(\Lambda(z)) = z + 1 + o(1)$, and $\ell_-(\Lambda(z)) = 1 + o(1)$. The left endpoint is symmetric. Integrating (2.15) gives

$$x(z) = m + b_+ + \frac{\lambda_+}{1 - \alpha_+} (z + 1)^{1 - \alpha_+} + o(1), \quad z \rightarrow +\infty, \quad (2.23)$$

$$x(z) = m + b_- - \frac{\lambda_-}{1 - \alpha_-} (1 - z)^{1 - \alpha_-} + o(1), \quad z \rightarrow -\infty, \quad (2.24)$$

for finite constants $b_{\pm}$. The additive constants do not affect the leading tail rates.

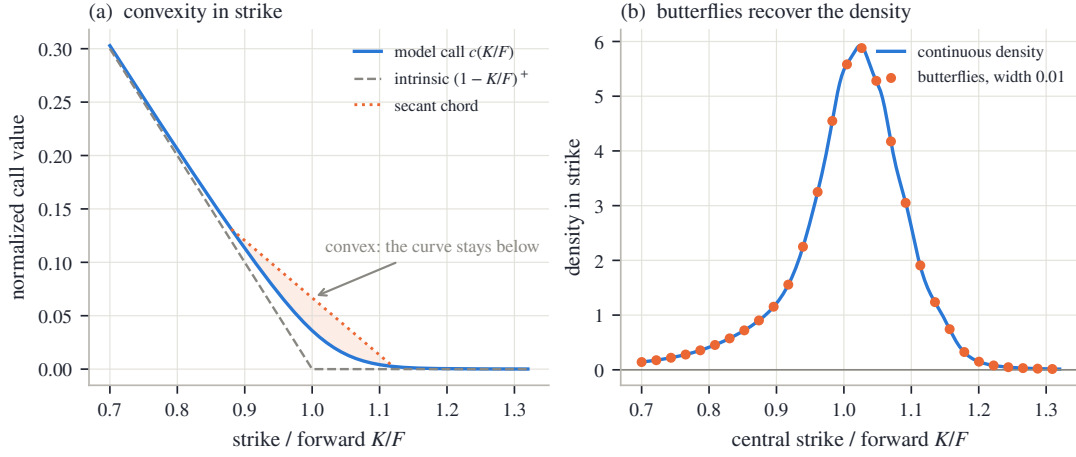


Figure 2.4: One-slice validity on the fitted SPY December 2026 law. Left: the call curve is convex in normalized strike. Right: symmetric butterfly prices, divided by squared half-width, converge to the density. The law supplies the convexity before calibration begins.

Proposition 2.5 (Log-return tail rates). *For a generalized LQD slice,*

$$-\log \mathbb{P}(X > x) \sim \left(\frac{(1 - \alpha_+)x}{\lambda_+} \right)^{1/(1-\alpha_+)}, \quad x \rightarrow +\infty, \quad (2.25)$$

$$-\log \mathbb{P}(X < -x) \sim \left(\frac{(1 - \alpha_-)x}{\lambda_-} \right)^{1/(1-\alpha_-)}, \quad x \rightarrow +\infty. \quad (2.26)$$

Proof. Invert (2.23) and use $\mathbb{P}(Z > z) = 1 - \Lambda(z) \sim e^{-z}$. The left proof is identical. $\square$

Write $p = 1/(1 - \alpha)$. The three cases are:

Tail exponent	Log-return tail	Extreme smile wing
$\alpha = 0$	$-\log \mathbb{P}(X > x) \asymp x$	$w(k) \asymp k $
$0 < \alpha < 1/2$	$-\log \mathbb{P}(X > x) \asymp x^p, 1 < p < 2$	$w(k)$ grows sublinearly
$\alpha = 1/2$	$-\log \mathbb{P}(X > x) \asymp x^2$	$w(k)$ tends to a constant

At $\alpha = 0$, $Y = e^X$ has a power tail. At $\alpha = 1/2$, a normal tail with standard deviation s corresponds to $\lambda = s/\sqrt{2}$. The two sides may use different exponents. Equity smiles often retain $\alpha_- = 0$ on the crash side and use either 0 or a small positive value on the right. The choice is a tail convention unless far-wing instruments identify it.

2.3.3 Moments and the finite-forward wall

Define the two moment boundaries

$$\bar{r}_{\pm} = \begin{cases} 1/\lambda_{\pm}, & \alpha_{\pm} = 0, \\ +\infty, & \alpha_{\pm} > 0. \end{cases} \quad (2.27)$$

Proposition 2.6 (Moment domain). *For every generalized LQD slice,*

$$\mathbb{E}[e^{rX}] < \infty \iff -\bar{r}_- < r < \bar{r}_+. \quad (2.28)$$

Thus the last finite positive moment of Y beyond its mean is $r_+^* = \bar{r}_+ - 1$, and the last finite inverse moment is $r_-^* = \bar{r}_-$, with $+\infty$ allowed.

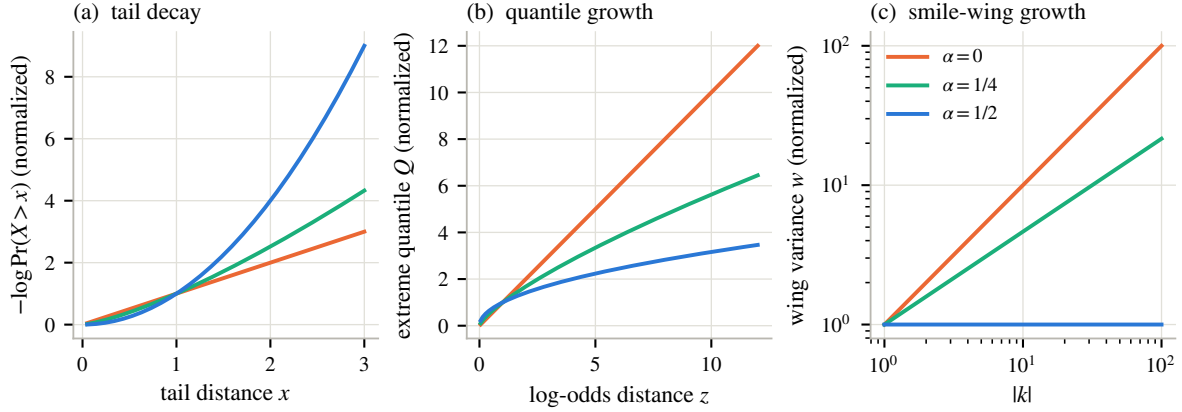


Figure 2.5: The tail exponent. Left: the leading negative log-tail grows like $x^{1/(1-\alpha)}$. Center: the extreme quantile grows like $z^{1-\alpha}$ in log-odds. Right: for $\alpha > 0$, total implied variance grows like $|k|^{(1-2\alpha)/(1-\alpha)}$; at $\alpha = 1/2$ it approaches a constant. Curves are normalized at one to compare rates, not levels.

Proof. Write the moment as

$$\mathbb{E}[e^{rX}] = \int_{-\infty}^{\infty} e^{rx(z)} \rho(z) dz.$$

If $\alpha_+ > 0$, then $x(z) = o(z)$ and the factor $\rho(z) \sim e^{-z}$ dominates every fixed positive r . If $\alpha_+ = 0$, the right integrand is asymptotic to a constant times $e^{(r\lambda_+-1)z}$. The left endpoint gives the other inequality in the same way. $\square$

The martingale condition is the single point $r = 1$ in this interval. When $\alpha_+ = 0$, it is available exactly when $\lambda_+ < 1$. A positive right tail exponent makes every positive moment finite; a positive left exponent makes every inverse moment finite. This is why two fits that are virtually indistinguishable on listed options can behave very differently for power payoffs, inverse products, and tail-sensitive risk measures.

2.3.4 From tail rate to smile wing

For exponential tails, Lee's moment formula gives the limiting slopes of total implied variance [12]:

$$\beta_R = \lim_{k \rightarrow +\infty} \frac{w(k)}{k}, \quad \beta_L = \lim_{k \rightarrow -\infty} \frac{w(k)}{|k|}. \quad (2.29)$$

Proposition 2.7 (Exponential-tail wing slopes). *If $\alpha_+ = 0$ and $\lambda_+ < 1$, and if $\alpha_- = 0$, then*

$$\beta_R = \frac{2\lambda_+}{(1 + \sqrt{1 - \lambda_+})^2}, \quad \beta_L = \frac{2\lambda_-}{(1 + \sqrt{1 + \lambda_-})^2}. \quad (2.30)$$

Both maps increase with the corresponding tail scale. The scale λ is not itself the smile slope: (2.30) is the exact conversion. For small λ , $\beta \sim \lambda/2$; as $\lambda_+ \uparrow 1$, $\beta_R \uparrow 2$.

Proof. Lee's function is $\Psi(r) = 2 - 4(\sqrt{r^2 + r} - r)$. Substitute $r_+^* = 1/\lambda_+ - 1$ and $r_-^* = 1/\lambda_-$ from proposition 2.6, then rationalize. The regular variation in proposition 2.5 turns Lee's limsup into a limit [1]. $\square$

Positive tail exponents have zero Lee slope because all moments are finite. Their sublinear wing growth remains explicit.

Proposition 2.8 (Intermediate and Gaussian-rate wings). *If $0 < \alpha_+ \leq 1/2$, then, as $k \rightarrow +\infty$,*

$$w(k) \sim \frac{1}{2} \left(\frac{\lambda_+}{1 - \alpha_+} \right)^{1/(1-\alpha_+)} k^{(1-2\alpha_+)/(1-\alpha_+)}. \quad (2.31)$$

If $0 < \alpha_- \leq 1/2$, the same formula holds on the left after replacing k by $|k|$ and the plus parameters by the minus parameters.

At $\alpha = 1/2$, total variance tends to $2\lambda^2$, the variance of the matching Gaussian tail. For $0 < \alpha < 1/2$, it diverges but does so sublinearly. The limit $\alpha \downarrow 0$ is nonuniform: every positive α gives zero Lee slope, while $\alpha = 0$ gives the positive slope (2.30). No finite strike strip can reliably distinguish zero from a sufficiently small positive exponent. Tail-exponent scenarios are therefore more stable than an unconstrained estimate for every expiry.

2.3.5 Expressiveness at a fixed tail class

The remaining question is whether the polynomial body imposes a material restriction once the tail class has been chosen. For fixed $\alpha_\pm$, define the residual log-speed of a differentiable target quantile Q^* by rearranging (2.11):

$$g^*(u) = \log Q^{*'}(u) + \log u + \log(1 - u) + \alpha_- \log \ell_-(u) + \alpha_+ \log \ell_+(u). \quad (2.32)$$

Proposition 2.9 (Approximation at fixed tail class). *Suppose g^* extends continuously to $[0, 1]$ and the target law has mean one. If $\alpha_+ = 0$, also suppose $e^{g^*(1)} < 1$. Then polynomials of the form (2.14) can be chosen so that $g_N \rightarrow g^*$ uniformly. The corresponding quantiles converge on compact rank intervals, and their call curves converge uniformly over all strikes.*

The proof is in appendix 2.A. The result says that increasing polynomial order can approximate any smooth law with the selected endpoint behavior; the tail skeleton, not the polynomial basis, is the substantive asymptotic restriction.

No finite-order slice has an atom, a zero-density interval, bounded support, or mass at zero. Those features require a quantile jump, a flat part, or a finite endpoint, none of which is available to a positive smooth transport. Multimodality is allowed: the density rises where the transport slows and falls where it accelerates. Increasing basis order permits more such changes in the body without altering the stated tail class.

2.4 Pricing from one cumulative ledger

Once the transport x is known, pricing does not require a new integration for every strike. One cumulative tail integral contains nearly every quantity required by a trader or a fitter.

2.4.1 The upper-share ledger

Definition 2.10 (Upper-share ledger). For an admissible slice define

$$G(z) = \int_z^\infty e^{x(t)} \rho(t) dt = \mathbb{E}[Y \mathbf{1}\{Z > z\}]. \quad (2.33)$$

The ordinary upper-tail probability is $1 - \Lambda(z)$. The ledger $G(z)$ weights the same outcomes by the terminal gross return $Y = e^X$. It is the fraction of the unit forward contributed by scenarios above rank $\Lambda(z)$. It satisfies

$$G(-\infty) = 1, \quad G(+\infty) = 0, \quad G'(z) = -e^{x(z)} \rho(z) < 0. \quad (2.34)$$

For a strike k , let z_k be the unique point at which the transported log-return reaches the strike, $x(z_k) = k$, and set $u_k = \Lambda(z_k) = F_X(k)$. Then $1 - u_k$ is the exercise probability.

Proposition 2.11 (Call, put, digital, and density). *For every admissible slice,*

$$c(k) = G(z_k) - e^k(1 - u_k), \quad (2.35)$$

$$P(k) = e^k u_k - (1 - G(z_k)), \quad (2.36)$$

$$c'(k) = -e^k(1 - u_k), \quad (2.37)$$

$$c''(k) = -e^k(1 - u_k) + e^k \frac{\rho(z_k)}{x'(z_k)}, \quad (2.38)$$

$$f_X(k) = e^{-k} \{c''(k) - c'(k)\} = \frac{\rho(z_k)}{x'(z_k)}. \quad (2.39)$$

Proof. The call is exercised on $Z > z_k$. Splitting its payoff into the received asset and the paid strike gives

$$c(k) = \mathbb{E}[Y \mathbf{1}\{Z > z_k\}] - e^k \mathbb{P}(Z > z_k) = G(z_k) - e^k(1 - u_k).$$

The put follows on the complementary event. When the call is differentiated, the moving-root term is

$$\{G'(z_k) + e^k \rho(z_k)\} \frac{dz_k}{dk} = 0$$

because $G'(z_k) = -e^{x(z_k)} \rho(z_k) = -e^k \rho(z_k)$. This gives (2.37). Since $dz_k/dk = 1/x'(z_k)$, one more derivative gives (2.38) and (2.39). $\square$

The pricing rule is worth retaining in words:

call value = upper asset share – strike $\times$ exercise probability.

(2.40)

The digital is $1 - u_k$. The density is one division of two quantities already present in the transport. Neither is obtained by differencing noisy option prices.

More generally, any integrable terminal payoff is a rank integral:

$$\mathbb{E}[H(Y)] = \int_{-\infty}^{\infty} H(e^{x(z)}) \rho(z) dz. \quad (2.41)$$

In particular, the log-contract variance is

$$w_{\text{vs}} = -2\mathbb{E}[\log Y] = -2 \int_{-\infty}^{\infty} x(z) \rho(z) dz. \quad (2.42)$$

This is the distribution-side form of the usual option-strip replication [4, 13]. Under continuous paths and continuous monitoring it is the fair variance-swap strike. Jumps and discrete sampling require their standard corrections.

2.4.2 A complete option calculation

In the constant-speed reference slice, choose $s = 0.081250$, which gives a six-month ATM volatility of twenty percent after exact repricing. The shift is $m = -0.010883$. For $k = 0.10$, the strike rank is $u_k = 79.65\%$. The ledger and cash legs are

$$G(z_k) = 0.247221, \quad e^k(1 - u_k) = 0.224876.$$

The normalized call is therefore

$$c(0.10) = 0.247221 - 0.224876 = 0.022345,$$

which corresponds to 20.60% implied volatility. Every market option uses the same three operations: locate the strike rank, read the ledger, and subtract the strike cash leg.

2.4.3 Reading level, skew, and curvature at the forward

This subsection is not needed for pricing. It explains exactly which local features of the fitted law generate the three familiar at-the-money (ATM) smile statistics. Let z_* solve $x(z_*) = 0$ and define

$$c_0 = G(z_*) - (1 - u_*), \quad u_* = \Lambda(z_*), \quad f_* = \frac{\rho(z_*)}{x'(z_*)}. \quad (2.43)$$

The ATM Black variance and half-standard-deviation are

$$w_0 = 4 \left[\Phi^{-1} \left(\frac{1 + c_0}{2} \right) \right]^2, \quad d = \frac{\sqrt{w_0}}{2}.$$

Define the digital gap

$$\Delta = u_* - \Phi(d). \quad (2.44)$$

Proposition 2.12 (Exact ATM handles). *The total-variance smile $B(k, w(k)) = c(k)$ satisfies*

$$w(0) = w_0, \quad w'(0) = \frac{2\sqrt{w_0}}{\varphi(d)} \Delta, \quad (2.45)$$

$$w''(0) = \frac{2\sqrt{w_0}}{\varphi(d)} f_* - 2 + \left(\frac{1}{2w_0} + \frac{1}{8} \right) w'(0)^2. \quad (2.46)$$

In volatility units, $\sigma(k) = \sqrt{w(k)}/\tau$,

$$\sigma(0) = \sqrt{w_0}/\tau, \quad \sigma'(0) = \frac{\Delta}{\varphi(d)\sqrt{\tau}}, \quad (2.47)$$

$$\sigma''(0) = \frac{1}{\sqrt{\tau}} \left[\frac{f_*}{\varphi(d)} - \frac{1}{\sqrt{w_0}} + \frac{\sqrt{w_0}}{4} \left(\frac{\Delta}{\varphi(d)} \right)^2 \right]. \quad (2.48)$$

The derivation is in section 2.A.2. The interpretation is simpler than the formulas:

- ATM option value fixes the volatility level;
- the digital gap Δ fixes the skew;
- density at the forward supplies the additional information needed for curvature.

For a matched Black law, $u_* = \Phi(d)$ and the skew is zero. A negative equity skew means $u_* < \Phi(d)$: the model assigns more probability above the forward than the flat-Black law with the same ATM call value.

2.4.4 The analytic calibration Jacobian

The call depends on a parameter in two ways: the ledger changes, and the rank at which the strike is reached moves. These effects appear to require a root sensitivity for every strike. In fact the moving-root term cancels exactly.

Theorem 2.13 (Envelope cancellation). *For every coordinate θ_j ,*

$$\frac{\partial c(k; \theta)}{\partial \theta_j} = \frac{\partial G(z; \theta)}{\partial \theta_j} \Big|_{z=z_k(\theta)}. \quad (2.49)$$

Proof. Differentiate (2.35). The coefficient of $\partial z_k / \partial \theta_j$ is $G'(z_k) + e^k \rho(z_k)$, which is zero by (2.34) and $x(z_k) = k$. $\square$

Let $b_j(u) = \partial g(u)/\partial \theta_j$ and $\bar{x} = x - m$. For fixed tail exponents,

$$\frac{\partial \bar{x}(z)}{\partial \theta_j} = \int_0^z x'(t) b_j(\Lambda(t)) dt, \quad (2.50)$$

$$\frac{\partial m}{\partial \theta_j} = - \int_{-\infty}^{\infty} e^{x(z)} \frac{\partial \bar{x}(z)}{\partial \theta_j} \rho(z) dz, \quad (2.51)$$

$$\frac{\partial G(z)}{\partial \theta_j} = \int_z^{\infty} e^{x(t)} \left(\frac{\partial m}{\partial \theta_j} + \frac{\partial \bar{x}(t)}{\partial \theta_j} \right) \rho(t) dt. \quad (2.52)$$

These are three cumulative passes: differentiate transport, differentiate the normalizing shift, then integrate the ledger sensitivity backwards. The denominator in the derivative of m is one because the accepted slice is already martingale-normalized. In raw coordinates,

$$b_L(u) = 1 - u, \quad b_R(u) = u, \quad b_{a_n}(u) = P_n(1 - 2u).$$

The same arrays supply the variance-swap sensitivity:

$$\frac{\partial w_{vs}}{\partial \theta_j} = -2 \int_{-\infty}^{\infty} \left(\frac{\partial m}{\partial \theta_j} + \frac{\partial \bar{x}(z)}{\partial \theta_j} \right) \rho(z) dz. \quad (2.53)$$

If a tail exponent is estimated, it adds the columns

$$b_{\alpha_-}(u) = -\log \ell_-(u), \quad b_{\alpha_+}(u) = -\log \ell_+(u) \quad (2.54)$$

to the derivative of $\log x'$. These columns grow in the remote ranks and are weakly identified by a finite strike strip. The reference policy fixes $\alpha_{\pm}$ and reports scenario fits instead.

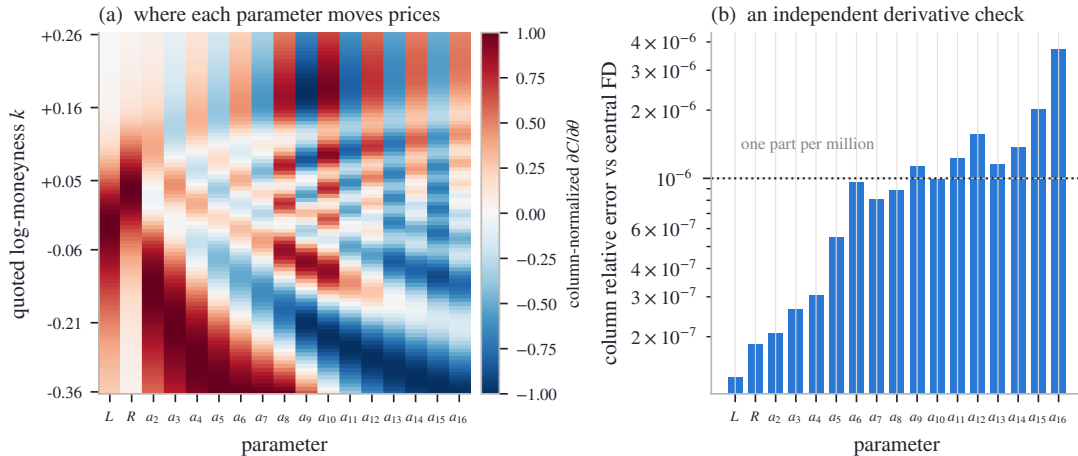


Figure 2.6: Analytic price sensitivities for the fitted SPY December 2026 slice. Left: one normalized sensitivity curve per parameter. Right: disagreement with an independent central difference. The comparison includes transport, normalization, ledger integration, root finding, and interpolation.

For $P = N + 1$ body and endpoint parameters and n_{grid} nodes, the entire sensitivity pass costs $O(Pn_{\text{grid}})$. Its cost is essentially independent of the number of quoted strikes. This is the practical value of the envelope cancellation: no slice rebuild and no repeated strike-root solve is needed for each finite-difference column.

2.5 Calibration

The coordinates make every trial slice a positive-density law. Calibration therefore has a narrower task: select, among those laws, one that explains the quotes without inventing unsupported detail between or beyond them. Four inputs govern that selection: the quoted price bands, basis order, smoothness penalty, and tail policy.

2.5.1 Use price residuals, measured in volatility units

Suppose the retained quotes are (k_i, σ_i) and $w_i = \sigma_i^2 \tau$. The basic one-slice objective is

$$\sum_i \left(\frac{c(k_i; \theta) - B(k_i, w_i)}{\partial_\sigma B(k_i, w_i) + \eta} \right)^2 + \omega_{\text{reg}} \sum_{n \geq 4} n^{2\nu} a_n^2. \quad (2.55)$$

Here $\partial_\sigma B = \varphi(d_+) \sqrt{\tau}$ is normalized Black vega, the first-order change in price for a one-unit change in volatility, and $\eta > 0$ is a vega floor. The numerator remains a price difference, so the optimizer never inverts an implied volatility. Dividing by vega merely expresses a central price error in the volatility units used by the market. In a remote wing, the floor prevents a nearly worthless option from receiving an unbounded weight.

The ridge starts at $n = 4$. The modes a_2 and a_3 carry the main curvature and asymmetry; higher modes describe progressively finer rank structure. The factor $n^{2\nu}$ makes fine oscillation increasingly expensive while leaving the leading smile shape largely data-driven.

An option quote is usually a bid–ask interval. If $[p_i^{\text{lo}}, p_i^{\text{hi}}]$ is the normalized price band and ϕ_i the vega normalizer, a band fit uses

$$\frac{(c_i - p_i^{\text{hi}})^+ + (p_i^{\text{lo}} - c_i)^+}{\phi_i}, \quad \sqrt{0.05} \frac{c_i - p_i^{\text{mid}}}{\phi_i}. \quad (2.56)$$

The first residual is zero inside the spread. The weak second term selects a stable center when many laws lie inside a wide band.

A volatility haircut h shrinks unreliable band edges toward the mid:

$$\sigma_i^{\text{lo}} = \min(\sigma_i^{\text{bid}} + h, \sigma_i^{\text{mid}}), \quad \sigma_i^{\text{hi}} = \max(\sigma_i^{\text{mid}}, \sigma_i^{\text{ask}} - h). \quad (2.57)$$

If the original spread is narrower than $2h$, the interval collapses to the mid. In the frozen sample this occurs for every featured SPY quote, while many wider NVDA bands remain active.

2.5.2 How many modes the strip can support

More modes always improve the best in-sample fit; they do not necessarily improve the inferred density. At order N the model has $N + 1$ body and endpoint parameters. A practical effective order is

$$N_{\text{eff}} = \max \left(\min \left(N, \left\lfloor \frac{n_q}{2} \right\rfloor - 1 \right), \min(N, 6) \right), \quad (2.58)$$

where n_q is the quote count. Away from the floor, this keeps roughly two quotes per shape degree of freedom. The rule controls identification; it is not a no-arbitrage condition. A larger order can change digitals and narrow butterflies between quotes while leaving vanilla residuals almost unchanged.

The quoted ranks make the same point directly. After a preliminary fit, record

$$[u_{\min}, u_{\max}] = \left[\min_i F_X(k_i), \max_i F_X(k_i) \right].$$

Modes acting mainly outside this interval are chosen by the ridge and the tail policy, not by quoted options. Reporting this interval states which part of the fitted distribution is directly informed by the market.

2.5.3 Optimization coordinates

Use the endpoint chart $(\log \lambda_-, \log \lambda_+, a_2, \dots, a_N)$ with the modes (2.19). Body moves then leave both tail coefficients fixed.

The right endpoint chart depends on the selected tail exponent:

$$\lambda_+ = \begin{cases} \text{expit } \xi, & \alpha_+ = 0, \\ e^\xi, & \alpha_+ > 0. \end{cases} \quad (2.59)$$

The first branch makes the finite-forward wall unreachable. The second is unconstrained because every positive moment exists. A separate moment prior may still cap λ_+ for numerical or risk purposes.

The raw coefficients L and R follow from the endpoint equations (2.18). This conversion is linear once the body coefficients are fixed. Analytic derivatives in the endpoint chart are obtained by multiplying the raw-coordinate Jacobian by this small chart Jacobian.

2.5.4 Choose the tail class outside the optimizer

The exponent α changes the asymptotic class, not merely the local wing slope. Over any finite strike range, a sufficiently small positive α can resemble $\alpha = 0$ while implying that every moment is finite and the Lee slope is zero. Listed vanilla options usually identify tail scale much more strongly than tail class. Unconstrained per-slice estimation is therefore ill-conditioned.

The implementation uses one of three policies:

1. $\alpha_- = \alpha_+ = 0$ for exponential log-return tails and power gross- return tails;
2. fixed exponents chosen for an asset class or product, with separate left and right values allowed;
3. a small grid of exponent scenarios, each recalibrated, with downstream prices reported across the scenarios.

The preferred practice is to calibrate a small number of stated tail scenarios and compare the downstream quantities that depend on them. Within one underlier, exponents are held common across expiry unless a monotone term-structure rule is imposed. This also stabilizes full-line calendar order; section 2.7 gives the exact restriction.

2.5.5 Starting values and solve sequence

For $\alpha_\pm = 0$, the constant-speed seed

$$s = \frac{\sqrt{3w_0}}{\pi} \quad (2.60)$$

matches log-return variance. It is about eight percent low in ATM implied variance at small w_0 , so it is only a starting value. For other tail exponents, set all body modes to zero and solve one scalar equation for the common endpoint level that matches the ATM call. This costs one-dimensional root finding and avoids an arbitrary variance proxy.

A complete slice fit follows six steps:

1. infer F and D , convert strikes to k_i , and convert bid, mid, and ask volatilities to normalized prices once;
2. fix (α_-, α_+) , select N_{eff} , and construct the residual rows and weights;
3. initialize the endpoint chart from the scalar reference slice or from the preceding expiry;
4. at each step, build (x, x', G, G') on one grid, price every quote with (2.35), and obtain the Jacobian from (2.50)–(2.52);
5. rebuild the accepted parameters on the publication grid and compute ATM handles, moments, wing asymptotics, variance-swap strike, and numerical certificates from that object;

6. for several expiries, replace independent sequential fits by the joint calendar-constrained solve of section 2.7.3.

Market conventions are fixed before optimization. Every trial slice remains a law. All reported quantities are read from the same final accepted object; there is no second smoothing or extrapolation layer that can alter its density, moments, or tails.

2.6 Numerical realization

The mathematical model lives on the whole log-odds line. A computer stores a finite central grid, so the two tails must be continued analytically. The important design rule is consistency: normalization, strike roots, prices, and calendar tests must all use the same continuation. One slice build then produces the transport, density ingredients, ledger, and sensitivity arrays.

2.6.1 The central grid

Use a symmetric grid $[-Z_{\max}, Z_{\max}]$. At each node compute $u = \Lambda(z)$, the two gauges (2.10), $g(u)$, and $x'(z)$ from (2.15). Cumulative quadrature anchored at $z = 0$ gives $\bar{x} = x - m$. The martingale integral gives m . Reverse cumulative quadrature gives G .

The reference sizes are 2001 nodes during optimization and 8001 nodes for the accepted slice, with $Z_{\max} = 40$. Cubic Hermite interpolation uses the exact nodal derivatives

$$x'(z) > 0, \quad G'(z) = -e^{x(z)}\rho(z) < 0. \quad (2.61)$$

A bracketed Hermite root followed by one Newton step gives z_k .

Three expressions need special numerical treatment. First,

$$\rho(z) = \text{expit}(z) \text{expit}(-z),$$

because $1 - \text{expit}(z)$ rounds to zero in the positive tail. Second, the cash leg is

$$e^k(1 - \Lambda(z_k)) = \exp\{k - \log(1 + e^{z_k})\},$$

evaluated with a log-sum-exp routine. Third, trial points are rejected before exponentiation if any grid value of g , x , or a sensitivity exponent exceeds the floating-point budget.

2.6.2 Tail continuation

For $z > Z_{\max}$, the leading transport continuation is

$$x_R(z) = x(Z_{\max}) + \frac{\lambda_+}{1 - \alpha_+} \left[(z + 1)^{1 - \alpha_+} - (Z_{\max} + 1)^{1 - \alpha_+} \right]. \quad (2.62)$$

The corresponding approximate root for $k > x(Z_{\max})$ is

$$z_R(k) + 1 = \left[(Z_{\max} + 1)^{1 - \alpha_+} + \frac{(1 - \alpha_+)(k - x(Z_{\max}))}{\lambda_+} \right]^{1/(1 - \alpha_+)}. \quad (2.63)$$

The formulas reduce to a straight line when $\alpha_+ = 0$. The left continuation is obtained by replacing $z + 1$ by $1 - z$ and reversing the sign of the transport increment.

In the exponential subclass, the leading missing martingale masses are

$$\int_{Z_{\max}}^{\infty} e^{\bar{x}} \rho \, dz \simeq \frac{e^{\bar{x}(Z_{\max}) - Z_{\max}}}{1 - \lambda_+}, \quad \int_{-\infty}^{-Z_{\max}} e^{\bar{x}} \rho \, dz \simeq \frac{e^{\bar{x}(-Z_{\max}) - Z_{\max}}}{1 + \lambda_-}. \quad (2.64)$$

For $\alpha_+ > 0$, the right correction is evaluated by one-dimensional log-domain quadrature under (2.62). Its endpoint approximation is

$$\int_{Z_{\max}}^{\infty} e^{x(z)} \rho(z) dz \simeq \frac{e^{x(Z_{\max}) - Z_{\max}}}{1 - x'(Z_{\max})}, \quad (2.65)$$

provided $x'(Z_{\max}) < 1$. The left denominator is $1 + x'(-Z_{\max})$.

The condition in (2.65) is numerical, not probabilistic. A positive tail exponent guarantees a finite forward, yet the normalizing integrand can first grow and only decay at an extremely remote rank. More precisely, if $x'(Z_{\max}) > 1$, it grows until the saddle $x'(z) = 1$. A production implementation must either locate that saddle and use scaled quadrature around it, or enforce the operational guard

$$x'(Z_{\max}) \leq 1 - \varepsilon_{\text{tail}}. \quad (2.66)$$

The reference implementation uses the guard. It excludes laws whose forward is finite in theory but numerically determined by ranks far beyond the grid.

For a strike beyond the right grid, evaluate the ledger under the same tail continuation and subtract the logistic cash leg. In the far limit,

$$c(k) \simeq e^{k - z_R(k)} \frac{x'(z_R(k))}{1 - x'(z_R(k))}. \quad (2.67)$$

At $\alpha_+ = 0$, this becomes $e^{k - z_R(k)} \lambda_+ / (1 - \lambda_+)$. Deep left calls are evaluated as a small put plus parity, avoiding subtraction of two numbers close to one.

2.6.3 Numerical certificates

The continuous formulas are arbitrage-free; a poorly resolved interpolation can still violate their monotonicity on a computer. Four independent checks are applied to every accepted slice.

1. On every cubic Hermite segment for x , verify the Fritsch–Carlson box

$$0 \leq d_0 \leq 3s, \quad 0 \leq d_1 \leq 3s,$$

where s is the secant slope and d_0, d_1 are endpoint derivatives. Apply the same check to $-G$ [6].

2. Across the retained strike range, compute analytic w, w', w'' and insert them in (2.5). Require $g_D \geq -\varepsilon_{\text{dens}}$, where the tolerance is stated in units of the local Black density factor.
3. Rebuild on a grid with four times as many nodes. Compare prices, densities, ATM handles, and the martingale integral at off-grid strikes and ranks.
4. Test tail formulas separately in the exponential logistic reference law and in a Gaussian-rate reference obtained from $X \sim N(-s^2/2, s^2)$ with $\alpha_{\pm} = 1/2$.

The Hermite box is sufficient rather than necessary. Its smallest remaining margin detects under-resolution before monotonicity is visibly lost.

The accepted parameter vector is rebuilt once on the publication grid. Prices, densities, ATM handles, variance-swap values, and calendar ledgers all read that same immutable object. Quote residuals measure market fit; basis order and tail exponents record modeling choices; numerical margins measure the reliability of the stored representation. They are reported separately.

2.7 Calendar order on the whole line

One-expiry validity removes butterfly arbitrage. A surface must also order expiries: moving expiry later cannot make a normalized call less valuable when the underlying forward-normalized price is a martingale. Because every LQD slice defines a price at every strike, the requirement applies on the whole strike line, not only where two expiries happen to have quotes.

2.7.1 The calendar inequality

Lemma 2.14 (Equal-log-moneyness calendar order). *Assume deterministic rates and carry. Let $F_{0,t}$ be the forward curve and $M_t = S_t/F_{0,t}$ a positive martingale. For $T_1 < T_2$ and every $k \in \mathbb{R}$,*

$$c_1(k) \leq c_2(k). \quad (2.68)$$

Proof. With $Y_i = M_{T_i}$, conditional Jensen gives

$$\begin{aligned} c_2(k) &= \mathbb{E}[(M_{T_2} - e^k)^+] \\ &\geq \mathbb{E}[\mathbb{E}[M_{T_2} \mid \mathcal{F}_{T_1}] - e^k]^+ = \mathbb{E}[(M_{T_1} - e^k)^+] = c_1(k). \end{aligned}$$

□

Equal k means equal strike divided by forward, so the cash strikes are $K_i = F_i e^k$ and are generally different. The two laws of Y_i have the same mean, while the farther law is more dispersed in the precise sense of convex order. If carry is stochastic, the numeraire and the objects compared must be specified before using this relation.

Black call price is strictly increasing in w at every finite k . Hence

$$c_1(k) \leq c_2(k) \iff w_1(k) \leq w_2(k). \quad (2.69)$$

Thus the visual desk rule for a stacked total-variance chart is exact: farther-expiry curves must lie above nearer-expiry curves at every log-moneyness.

2.7.2 Move the calendar test from strike to rank

Calls from two expiries reach a given strike at different ranks. Directly comparing them therefore mixes the two transport maps. The ledger places all expiries on the same probability-rank axis. The following conjugacy shows that no information is lost by making this change.

Lemma 2.15 (Ledger-call conjugacy). *For every admissible slice,*

$$\begin{aligned} c(k) &= \max_{z \in [-\infty, \infty]} \{G(z) - e^k(1 - \Lambda(z))\}, \\ G(z) &= \min_{k \in [-\infty, \infty]} \{c(k) + e^k(1 - \Lambda(z))\}. \end{aligned} \quad (2.70)$$

The optimizers are $z = z_k$ and $k = x(z)$.

Proof. For fixed k ,

$$G(z) - e^k(1 - \Lambda(z)) = \int_z^\infty (e^{x(t)} - e^k) \rho(t) dt.$$

The integrand is negative before z_k and positive after it. Starting the integral at z_k maximizes the expression and gives the call. The first identity implies $c(k) + e^k(1 - \Lambda(z)) \geq G(z)$, with equality at $k = x(z)$, which gives the second identity. □

Theorem 2.16 (Calendar order is ledger order). *For two admissible mean-one slices,*

$$c_1(k) \leq c_2(k) \text{ for every } k \in \mathbb{R} \iff G_1(z) \leq G_2(z) \text{ for every } z \in \mathbb{R}. \quad (2.71)$$

Proof. If $G_1 \leq G_2$, take maxima in the first line of (2.70). If $c_1 \leq c_2$, take minima in the second line. $\square$

The theorem says that checking all strikes is exactly the same as checking all ranks. The ledger axis is common across expiries and the slice builder already stores it, making it the natural coordinate for both calibration constraints and the final certificate. Ordered marginals are compatible with at least one martingale by the theorems of Strassen and Kellerer [11, 16]. The marginal surface does not select that martingale dynamics.

2.7.3 Full-line constraints in calibration

A constraint restricted to the common quoted span controls an interpolant there; it says nothing about the published extrapolation. That restriction is appropriate only for a model deliberately left undefined outside the quoted span. An LQD slice is defined on all of $\mathbb{R}$, so the accepted maturity stack imposes

$$G_{j+1}(z) - G_j(z) \geq 0 \quad \text{for every } z \in \mathbb{R} \quad (2.72)$$

for every adjacent pair of expiries.

The expiries are fitted jointly because changing either member of an adjacent pair changes the calendar gap. The data term is the sum of the slice objectives. The inequalities (2.72) are hard constraints, not residuals whose violation can be traded against a slightly better quote fit. Standard constrained solvers use the analytic ledger sensitivities from (2.52); a least-squares implementation may use the active-set exchange described below. In either case, acceptance requires the full-line certificate.

Global control necessarily couples unquoted wings across maturity because those wings belong to the published surface. Choose the tail policy for the stack before fitting. With common tail exponents, eventual calendar order requires

$$\lambda_{-,j+1} \geq \lambda_{-,j}, \quad \lambda_{+,j+1} \geq \lambda_{+,j}, \quad (2.73)$$

apart from exact ties, where the next asymptotic constant decides the order. These inequalities are imposed in the endpoint chart. If exponents vary by expiry, the farther law cannot have a lighter asymptotic tail: on either side $\alpha_{\pm,j+1} > \alpha_{\pm,j}$ forces a calendar reversal sufficiently far out. Common $\alpha_{\pm}$ across a maturity stack is the simpler policy.

If the hard constraints are inconsistent with the quote bands, no law stack can satisfy both those bands and the selected tail policy. The meaningful diagnostic is the smallest quote-band relaxation needed for feasibility. Shrinking the calendar domain would merely hide the inconsistency outside the chosen interval.

2.7.4 A complete calendar certificate

The all-line constraint still appears infinite-dimensional. The transport geometry reduces it to finitely many evaluations. Let

$$H(z) = G_2(z) - G_1(z).$$

At both endpoints H tends to zero. Its derivative has a simple sign:

$$H'(z) = \rho(z)\{e^{x_1(z)} - e^{x_2(z)}\}. \quad (2.74)$$

Thus the ledger gap can turn only where the two quantile curves cross. Between two consecutive crossings it is monotone. Every possible negative minimum therefore occurs at a crossing or in an analytic tail.

This gives a finite certificate for the stored surface:

1. isolate every root of $x_{j+1} - x_j$ on the central grid; for cubic Hermite transports this is a cubic root problem on each segment;
2. evaluate $G_{j+1} - G_j$ at those roots and at the two grid boundaries;
3. continue both transports with (2.62) and its left analogue, isolate any tail roots, and evaluate the ledger gap there;
4. check the limiting tail order from the exponents, endpoint coefficients, and, in a tie, the asymptotic constants.

Nonnegative values at all these candidate minima prove (2.72) on the whole real line. This is a finite certificate of a continuum property; no arbitrary strike cutoff is involved.

During calibration, the current violating minimum is added to the active constraint set. This exchange step is repeated until the certificate finds no negative minimum. In practice the active ranks are few because adjacent smooth quantile curves cross only a few times.

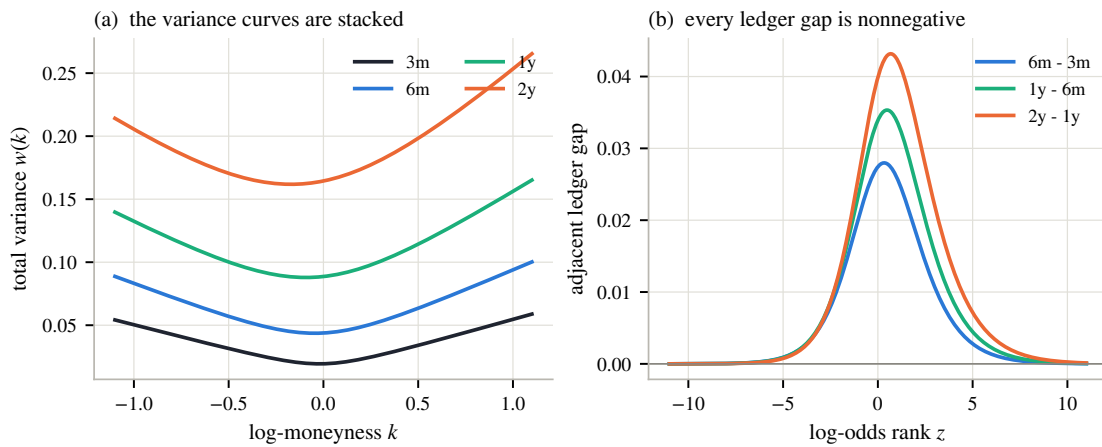


Figure 2.7: A globally ordered synthetic LQD maturity stack. Left: total variance at four expiries; the curves do not cross. Right: adjacent ledger gaps on the common rank axis; each remains nonnegative and returns to zero at both endpoints. The displayed curves illustrate the order; the quantile-root test certifies it on the full line.

At extreme strikes, both Black price and vega can underflow while the implied variance remains finite. A publication chart therefore evaluates remote wings from (2.30) or (2.31), matched to the central inversion, rather than inverting numerically zero prices. The chart then displays the globally certified surface instead of numerical noise from an ill-conditioned inverse.

2.8 What the model reveals in practice

The examples answer three practical questions: how local skew arises from the law, what an endpoint coefficient says about unquoted risk, and why a close vanilla fit does not identify a fine-grained density. They use frozen SPY and NVDA option chains from the 2026-08-03 US session, with eight expiries per underlier. The fits use the exponential-tail subclass $\alpha_- = \alpha_+ = 0$, order $N = 16$ subject to (2.58), haircut bands, and ridge $\omega_{\text{reg}} = 10^{-6}$ with $\nu = 1$. The examples therefore test the original LQD subclass; the generalized tail exponent is evaluated by separate scenario fits.

2.8.1 A liquid index: skew is a digital imbalance

The slice contains 94 quotes and fits them to 3.3 volatility basis points rms. The exact ATM formulas give volatility 14.76%, skew -0.430 , and curvature 4.40. The forward rank is 41.31%.

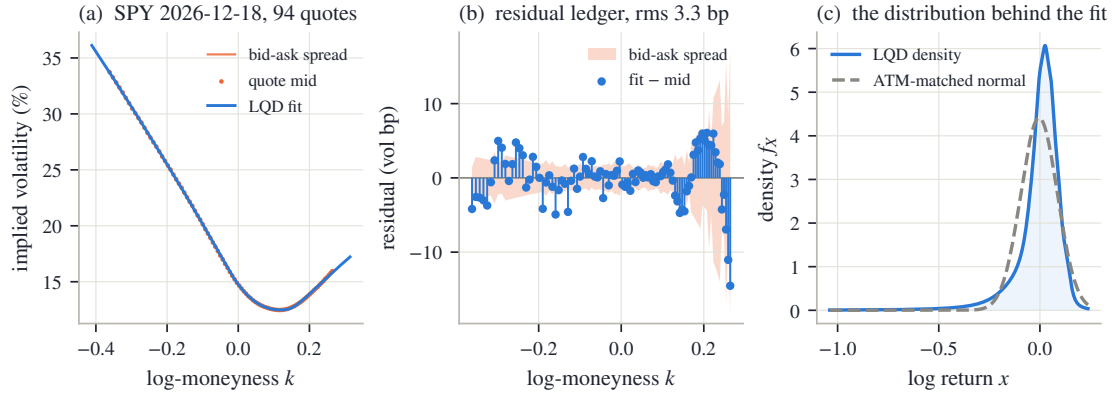


Figure 2.8: The SPY December 2026 slice. Left: quoted volatilities and the fitted smile. Center: signed residuals in volatility basis points. Right: the fitted density and an ATM-matched normal density. The left-shifted mass and richer model digital produce the negative equity skew.

below the matched Black rank, so $\Delta < 0$ and (2.47) gives a negative skew. The log-contract variance corresponds to 19.84% annualized volatility.

For the listed 800 call, $k = 0.0409$ and the strike rank is 65.27%. The asset-share and cash entries are 0.378366 and 0.361834, hence

$$c(0.0409) = 0.378366 - 0.361834 = 0.016532.$$

This is \$12.70 per share and corresponds to 13.37% implied volatility.

The endpoint coefficients of the exponential subclass are $\lambda_- = 0.432$ and $\lambda_+ = 0.066$. They imply Lee slopes $\beta_L = 0.179$ and $\beta_R = 0.034$. These are asymptotic properties of the fitted continuation, not measurements at the outer quoted strikes.

2.8.2 A single name: vanilla fit does not settle tail policy

The long NVDA slice has endpoint coefficients $\lambda_- = 1.421$ and $\lambda_+ = 0.225$. Its left coefficient exceeds one. This is admissible because negative log-returns do not threaten the forward; it implies that $\mathbb{E}[Y^{-1}]$ is infinite in the exponential-tail subclass. A scenario with $\alpha_- > 0$ keeps the fitted central law but makes all inverse moments finite. Prices of inverse-sensitive or crash-tail positions distinguish the two policies more directly than vanilla fit error.

On the shortest NVDA expiry, 17 retained quotes give $N_{\text{eff}} = 7$. The order guard prevents a sparse strip from assigning an independent high-frequency shape mode to each quote.

2.8.3 Basis order: the smile may agree while the density differs

The synthetic benchmark is a martingale-normalized mixture of two lognormal components. Its density is bimodal, while its implied-volatility curve has only a mild shoulder.

At order 16 the L^1 density error is 0.047; at order 6 it is 0.108. Yet the implied-volatility curves differ by at most 41.8 volatility basis points on the comparison grid. Basis order should therefore be selected for the downstream use. Broad vanilla prices tolerate a lower order; digitals and narrow butterflies require a finer rank representation and stronger identification checks.

2.8.4 Relation to other smile models

Breeden–Litzenberger identify the call curve with a density [3]. Quantile modeling supplies the rank coordinate [9, 14]; Petersen–Müller use a log-quantile-density transform for density-valued data [15]; the metalog family is a related direct quantile expansion [10]. Lee and Benaim–Friz connect moments to implied-variance wings [1, 12].

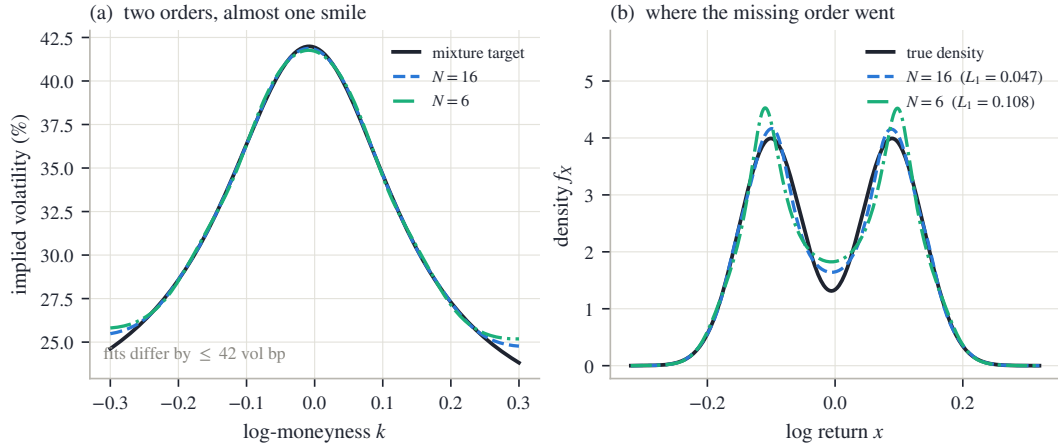


Figure 2.9: The same bimodal target fitted at orders 6 and 16. Left: the two implied-volatility curves nearly coincide over the quoted range. Right: the densities differ; the lower-order model shifts the modes and fills part of the trough. Vanilla residuals alone do not identify fine density structure.

SVI and SSVI specify total variance directly and characterize admissible parameter regions [7, 8]. Constrained price smoothers work directly with convex call curves [5]. LQD instead makes the law primary. Its main computational object is the pair (x, G) : the transport supplies density and the ledger supplies prices and calendar order.

2.8.5 What follows, what is fitted, and what is chosen

Derived	Fitted and checked	Chosen
Positive density, unit mass, unit forward, decreasing convex calls, and butterfly-free prices Tail rate (2.25), moment domain (2.28), Lee slopes or sublinear wing law	Quote residuals, interpolation monotonicity, Durrleman margin, and grid convergence Tail-quadrature accuracy, operational tail guard, and scenario sensitivity beyond quoted ranks	Basis order, ridge, vega floor, quote-band rule, and numerical tolerances Tail exponents $\alpha_{\pm}$ and any additional endpoint-moment limits
Prices, digitals, density, ATM handles, log-contract variance, and analytic Jacobian from (x, G)	Multi-start stability, active calendar constraints, and the full-line quantile-root certificate	Common or monotone tail policy across expiry and the permitted quote-band relaxation
Full-line call order, ledger order, and noncrossing total variance are equivalent	The accepted maturity stack satisfies the order to its stated normalized-price tolerance	Dynamics between the fitted marginal laws are not selected by the surface

Table 2.1: The LQD operating contract. The first column follows from the model; the second is measured on the implementation; the third is supplied before calibration.

A fitted surface is published with its quote residuals, effective orders, quoted-rank spans, tail exponents and coefficients, moment limits, numerical margins, and minimum adjacent ledger gap. These quantities are sufficient to reconstruct how the surface is fitted, extrapolated, and certified.

2.A Proofs and asymptotic calculations

2.A.1 The volatility-space density factor

Let $c(k) = B(k, w(k))$. The identity $e^k \varphi(d_-) = \varphi(d_+)$ gives

$$\partial_k B = -e^k \Phi(d_-), \quad \partial_w B = \frac{\varphi(d_+)}{2\sqrt{w}}.$$

The required second partial derivatives are

$$\begin{aligned} \partial_{kk} B &= -e^k \Phi(d_-) + \frac{\varphi(d_+)}{\sqrt{w}}, \\ \partial_{kw} B &= \frac{d_+ \varphi(d_+)}{2w}, \\ \partial_{ww} B &= -\frac{d_+ \varphi(d_+)}{2\sqrt{w}} \partial_w d_+ - \frac{\varphi(d_+)}{4w^{3/2}}, \quad \partial_w d_+ = \frac{k}{2w^{3/2}} + \frac{1}{4\sqrt{w}}. \end{aligned}$$

Twice differentiating $c(k)$ and subtracting the first derivative gives

$$c'' - c' = (\partial_{kk} B - \partial_k B) + (2\partial_{kw} B - \partial_w B)w' + \partial_{ww} B(w')^2 + \partial_w B w''.$$

Now

$$2\partial_{kw} B - \partial_w B = -\frac{k\varphi(d_+)}{w^{3/2}}, \quad d_+ \partial_w d_+ = -\frac{k^2}{2w^2} + \frac{1}{8}.$$

Factoring $\varphi(d_+)/\sqrt{w}$ yields

$$c'' - c' = \frac{\varphi(d_+)}{\sqrt{w}} \left[\left(1 - \frac{kw'}{2w}\right)^2 - \frac{(w')^2}{4} \left(\frac{1}{w} + \frac{1}{4}\right) + \frac{w''}{2} \right].$$

Finally $f_X(k) = e^{-k}(c'' - c')$ and $e^{-k}\varphi(d_+) = \varphi(d_-)$, which gives (2.5).

2.A.2 The ATM identities

At $(k, w) = (0, w_0)$, with $d = \sqrt{w_0}/2$,

$$\begin{aligned} \partial_k B &= -(1 - \Phi(d)), & \partial_w B &= \frac{\varphi(d)}{2\sqrt{w_0}}, \\ \partial_{kk} B &= -(1 - \Phi(d)) + \frac{\varphi(d)}{\sqrt{w_0}}, & \partial_{kw} B &= \frac{\varphi(d)}{4\sqrt{w_0}}, \\ \partial_{ww} B &= -\frac{\varphi(d)}{4w_0^{3/2}} - \frac{\varphi(d)}{16\sqrt{w_0}}. \end{aligned} \tag{2.75}$$

From (2.37)–(2.39),

$$c'(0) = -(1 - u_*), \quad c''(0) = f_* - (1 - u_*).$$

One derivative of $B(k, w(k)) = c(k)$ gives

$$w'(0) = \frac{-(1 - u_*) + (1 - \Phi(d))}{\varphi(d)/(2\sqrt{w_0})} = \frac{2\sqrt{w_0}}{\varphi(d)} \Delta.$$

Two derivatives give

$$\partial_{kk} B + 2\partial_{kw} B w' + \partial_{ww} B (w')^2 + \partial_w B w'' = c''.$$

Substitution of (2.75) and the first-derivative formula reduces this to (2.46). Finally,

$$\sigma' = \frac{w'}{2\sqrt{\tau w}}, \quad \sigma'' = \frac{w''}{2\sqrt{\tau w}} - \frac{(w')^2}{4\sqrt{\tau} w^{3/2}},$$

which gives (2.47)–(2.48).

2.A.3 Sublinear wing asymptotics

Consider the right wing with $0 < \alpha_+ \leq 1/2$ and set

$$p = \frac{1}{1 - \alpha_+}, \quad A = \left(\frac{1 - \alpha_+}{\lambda_+} \right)^p.$$

Proposition 2.5 gives $-\log \mathbb{P}(X > x) \sim Ax^p$. The call has the tail-integral form

$$c(k) = \int_k^\infty e^x \mathbb{P}(X > x) dx.$$

Since $p > 1$, endpoint Laplace bounds give

$$-\log c(k) \sim Ak^p. \quad (2.76)$$

For example, for every $\varepsilon > 0$ and all large x , $e^{-(A+\varepsilon)x^p} \leq \mathbb{P}(X > x) \leq e^{-(A-\varepsilon)x^p}$; integrating the two bounds and then sending ε to zero proves (2.76).

All positive moments exist, so Lee's formula gives $w(k)/k \rightarrow 0$. Mills' ratio in the Black formula then gives

$$-\log B(k, w(k)) \sim \frac{k^2}{2w(k)}. \quad (2.77)$$

Equating (2.76) and (2.77) yields

$$w(k) \sim \frac{1}{2A} k^{2-p},$$

which is (2.31). The left proof uses the put and is identical.

2.A.4 Approximation at fixed tail class

Let g^* satisfy proposition 2.9. By Weierstrass' theorem there are polynomials g_N with $\varepsilon_N = \|g_N - g^*\|_\infty \rightarrow 0$. The endpoint chord and $P_2, \dots, P_N$ span every polynomial of degree at most N .

The speed ratio is bounded uniformly on the whole line:

$$e^{-\varepsilon_N} \leq \frac{x'_N(z)}{x^{*'}(z)} \leq e^{\varepsilon_N}.$$

Consequently $\bar{x}_N \rightarrow \bar{x}^*$ uniformly on every compact z interval. It remains to control the tails and the martingale shifts.

If $\alpha_+ > 0$, boundedness of the g_N gives constants C_1, C_2 such that

$$\bar{x}_N(z) \leq C_1 + C_2 z^{1-\alpha_+}, \quad z \geq 0.$$

Hence $e^{\bar{x}_N(z)} \rho(z)$ is dominated by an integrable multiple of $\exp(C_2 z^{1-\alpha_+} - z)$. If $\alpha_+ = 0$, the strict endpoint margin $e^{g^*(1)} < 1$ supplies $\bar{\lambda} < 1$ and a common bound $e^{\bar{x}_N(z)} \rho(z) \leq C e^{-(1-\bar{\lambda})z}$ for large z . On the left, $\bar{x}_N(z) \leq 0$ and $\rho(z) \leq e^z$. Dominated convergence therefore gives convergence of the normalizers and shifts.

The quantiles converge at every fixed rank. The same envelopes provide a common moment of order strictly larger than one, so the gross returns are uniformly integrable. Under the common uniform-rank coupling,

$$\sup_k |c_N(k) - c^*(k)| \leq \mathbb{E} |e^{Q_N(U)} - e^{Q^*(U)}| \rightarrow 0.$$

This proves uniform convergence of call prices.

2.A.5 Tail restrictions imposed by calendar order

On the right, (2.23) and endpoint integration give

$$\log G(z) = -z + \frac{\lambda_+}{1 - \alpha_+} z^{1-\alpha_+} + o(z^{1-\alpha_+}).$$

If a farther expiry has a larger α_+ , its positive correction has a smaller power and its ledger is eventually below the nearer ledger. If the exponents are equal, the larger λ_+ gives the larger eventual ledger. The left lower-share ledger satisfies

$$\log(1 - G(z)) = z - \frac{\lambda_-}{1 - \alpha_-} |z|^{1-\alpha_-} + o(|z|^{1-\alpha_-}), \quad z \rightarrow -\infty.$$

The same comparison gives the left conditions stated after (2.73). Equal exponents and equal endpoint coefficients leave the additive asymptotic constants to decide the order.

2.B Implementation blueprint

The listings record the numerical core. They omit only ordinary exception handling and the choice of quadrature library.

2.B.1 Stable grid quantities

The endpoint gauges should be evaluated from log-odds, not from rounded ranks:

$$\ell_-(\Lambda(z)) = 1 + \log(1 + e^{-z}), \quad \ell_+(\Lambda(z)) = 1 + \log(1 + e^z). \quad (2.78)$$

Both logarithms use a softplus routine. This remains accurate when $\Lambda(z)$ has rounded to zero or one.

Listing 2.1: Build one generalized LQD slice.

```
def build(theta, alpha, grid):
    # Endpoint chart: log lambda-, right-tail coordinate, body modes.
    lam_m, lam_p, a = decode_endpoint_chart(theta, alpha)
    L, R = raw_chord(lam_m, lam_p, a)

    u = expit(grid.z)
    ell_m = 1 + softplus(-grid.z)
    ell_p = 1 + softplus( grid.z)
    rho = expit(grid.z) * expit(-grid.z)

    g = (1-u)*L + u*R + a @ grid.Pmat
    log_dx = g - alpha.minus*log(ell_m) \
              - alpha.plus *log(ell_p)
    check_exponent_budget(log_dx)
    dx = exp(log_dx)

    if alpha.plus == 0 and lam_p >= 1 - wall_eps:
        raise InadmissibleForward
    if dx[-1] > 1 - tail_eps:
        raise RemoteSaddleBeyondGrid

    xbar = cumulative_simpson(dx, anchor=grid.mid)
    check_exponent_budget(xbar)

    I = integrate(exp(xbar)*rho)
    I += left_tail_mass(xbar[0], theta, alpha, grid)
    I += right_tail_mass(xbar[-1], theta, alpha, grid)
```

```

m = -log(I)
x = m + xbar

dG = -exp(x)*rho
G = reverse_integrate(-dG)
G += right_tail_mass(x[-1], theta, alpha, grid)
return Slice(x=x, dx=dx, G=G, dG=dG,
             m=m, lam=(lam_m, lam_p),
             alpha=alpha, theta=theta)

```

The two tail-mass functions use the continuation (2.62) and its left counterpart. For $\alpha = 0$ they use (2.64). For $\alpha > 0$ they integrate in log space after the change $z = Z_{\max} + t$. All tail corrections and their parameter derivatives enter both the primal build and the sensitivity pass.

The median anchor fixes $\bar{x}(0) = 0$. Simpson quadrature is fourth order on the uniform grid. Exact nodal derivatives make the cubic Hermite representation fourth order off grid without a global spline solve.

2.B.2 Call evaluation

Listing 2.2: Evaluate one normalized call.

```

def call_price(slc, k, grid):
    if k > slc.x[-1]:
        z = right_tail_root(slc, k, grid) # analytic tail continuation
        share = right_ledger_tail(slc, z, grid)
        cash = exp(k - softplus(z))
        return share - cash

    if k < slc.x[0]:
        put = left_put_tail(slc, k, grid)
        return 1 - exp(k) + put

    z = bracketed_hermite_root(slc.x, slc.dx, k)
    share = hermite_eval(slc.G, slc.dG, z)
    cash = exp(k - softplus(z))
    return share - cash

```

The right branch uses the same tail law as the martingale normalizer. A separate volatility extrapolation would generally break the density and calendar identities.

2.B.3 Sensitivity pass

Listing 2.3: All fixed-tail parameter sensitivities.

```

def sensitivities(slc, grid):
    # Rows are derivatives of g in the chosen coefficient chart.
    B = basis_columns_in_chart(slc.theta, grid.u)
    dxbar = cumulative_simpson(slc.dx * B,
                              anchor=grid.mid, axis=1)
    weight = exp(slc.x) * grid.rho
    dm = -integrate(weight * dxbar, axis=1)
    add_tail_derivatives_to_shift(dm, slc, grid)
    dx = dm[:, None] + dxbar

    integrand = weight[None, :] * dx
    dG = reverse_integrate(integrand, axis=1)
    add_tail_derivatives_to_ledger(dG, slc, grid)
    dG_slope = -integrand
    dvs = -2 * integrate(dx * grid.rho, axis=1)
    return Sensitivities(dG=dG, dG_slope=dG_slope, dvs=dvs)

```

```
def price_jacobian_row(slc, sens, k):
    z = strike_root(slc, k)
    return hermite_eval(sens.dG, sens.dG_slope, z)
```

The martingale derivative is $\partial m = -\int e^x \partial \bar{x} \rho dz$, including both tail pieces. Envelope cancellation then turns the ledger sensitivity at the strike root directly into the call sensitivity.

2.B.4 Full-line calendar constraints

During the joint maturity fit, maintain a small active set of ranks for every adjacent expiry pair.

Listing 2.4: Exchange algorithm for global calendar order.

```
def fit_surface(nodes):
    active = initial_calendar_ranks(nodes)
    while True:
        surface = constrained_joint_solve(nodes, active)
        worst = full_line_calendar_minima(surface)
        if worst.value >= -calendar_tol:
            return surface
        active[worst.pair].add(worst.rank)

def full_line_calendar_minima(surface):
    out = []
    for near, far in adjacent_pairs(surface):
        roots = all_piecewise_cubic_roots(far.x - near.x)
        roots += analytic_tail_roots(far, near)
        points = [-inf] + roots + [inf]
        gaps = [0.0] \
            + [far.ledger(z) - near.ledger(z) for z in roots] \
            + [0.0]
        out.append(min_with_location(gaps, points))
    return min(out)
```

If the difference of the two cubic transports vanishes on an entire segment, the ledger gap is constant there by (2.74); either endpoint represents the segment. Tail root isolation uses (2.62). The final certificate is rerun on the publication grid and with tighter tail quadrature than the optimizer uses.

2.B.5 Failure tests

The implementation treats the following states separately.

1. *Rank saturation*: use (2.78), two-sided logistic factors, and log-space cash legs.
2. *Interior overflow*: inspect the full arrays of g , $\log x'$, and $\bar{x}$ before exponentiation; admissible endpoints do not bound body modes.
3. *Exponential-tail wall*: for $\alpha_+ = 0$, keep $\lambda_+ \leq 1 - \varepsilon_{\text{wall}}$ in a logistic chart.
4. *Remote light-tail saddle*: for $\alpha_+ > 0$, apply (2.66) or use a saddle-centered quadrature.
5. *Near-zero option prices*: compare calls and calendars in log space and obtain plotted remote implied variance from the analytic wing law.
6. *Calendar gaps*: a finite penalty is not an acceptance test; the quantile-root certificate must pass for every adjacent expiry pair.

2.B.6 Reference defaults

Quantity	Default	Quantity	Default
Basis order N	16 (cap 24)	Order guard	(2.58)
Grid	$[-40, 40]$	Nodes	8001 (fit: 2001)
Tail exponents	stated scenarios	Exponential wall	$1 - 10^{-6}$
Remote-saddle margin	10^{-4}	Exponent budget	700
Ridge $(\omega_{\text{reg}}, \nu)$	$(10^{-6}, 1)$	Vega floor η	10^{-4}
Haircut h	0.005 volatility	Solver	constrained trust region
Slice check	801 strikes	Calendar check	all quantile crossings
Optimization tolerance	10^{-10}	Publication rebuild	fourfold check

Table 2.2: Reference numerical defaults. Tail exponents and their maturity policy are model inputs and are therefore reported with the fit.

The shortest porting sequence is: reproduce the logistic reference law; reproduce a normal law with $\alpha_{\pm} = 1/2$; implement the stable cash leg and both tail continuations; add the analytic sensitivity pass; then add the one-slice and full-line calendar certificates.

2.C Data and reproducibility

Every empirical number and inherited market figure in this alternative chapter comes from the frozen JSON snapshot used by the book. It contains SPY and NVDA option chains for eight expiries each from the 2026-08-03 US session, together with forwards, discount factors, retained quotes, fitted coefficients, and display curves. Reproduction requires no live market-data connection.

The stored fits use $\alpha_- = \alpha_+ = 0$. They are rebuilt from the frozen coefficients rather than refitted, so their prices and all macros remain those of the published snapshot. The two new figures in this version are synthetic: fig. 2.5 plots the derived asymptotic rates, and fig. 2.7 plots an exactly specified constant-speed LQD maturity stack. Neither introduces market observations.

Tail-policy comparisons recalibrate every exponent scenario under the same quotes and joint calendar constraints. Changing $\alpha_{\pm}$ at fixed body coefficients is only a local sensitivity.

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